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4.3 Particle Model of Matter

AQA GCSE Physics (8463) — Paper 1

4.3.1 Density

Density is the mass of a substance per unit volume. Symbol: `ρ` (rho).

`ρ = m / V`

where `ρ` = density (kg/m³ or g/cm³), `m` = mass (kg or g), `V` = volume (m³ or cm³).

Density tells us how tightly packed the particles of a substance are. A denser material has more mass squeezed into the same volume. Water has a density of 1000 kg/m³; iron is about 8 times denser at 7874 kg/m³, which is why iron sinks in water. Air at sea level has a density of only 1.2 kg/m³, which is why balloons filled with helium (0.18 kg/m³) float upward.

Unit Conversion

Density is commonly expressed in two units:

- **kg/m³** — SI unit (used for most calculations)
- **g/cm³**

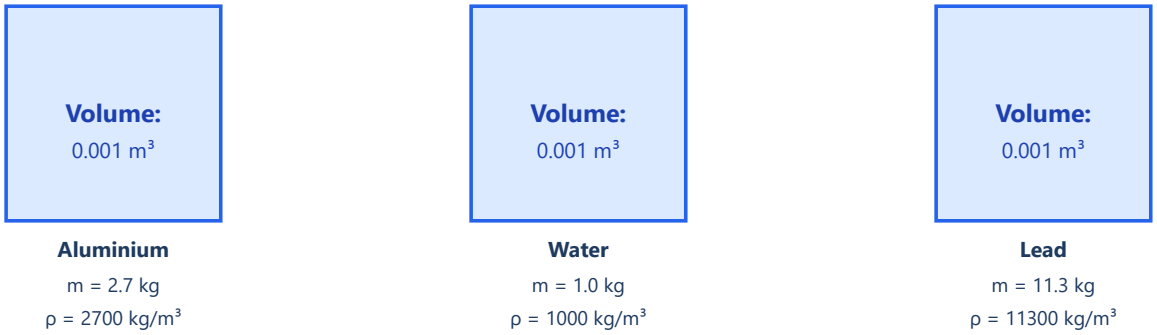
Conversion: `1 g/cm³ = 1000 kg/m³`

Example: The density of water is 1.0 g/cm³, which equals 1000 kg/m³. Aluminium is 2.7 g/cm³, which equals 2700 kg/m³.

Why Objects Float or Sink

An object floats in a fluid if its density is less than the density of that fluid. An object sinks if its density is greater than the fluid's density. This is why oil (density ≈ 900 kg/m³) floats on water (1000 kg/m³), but iron (7874 kg/m³) sinks. Even though iron is much denser, a steel ship with many air-filled chambers has an average density less than water, so it floats.

Density: Same volume, different masses (and therefore different densities)



Common Densities (at room temperature and 1 atm pressure)

Material	Density (kg/m³)	Density (g/cm³)
Air	1.2	0.0012
Wood (pine)	~700	~0.7
Water	1000	1.0
Aluminium	2710	2.71
Iron	7874	7.87
Lead	11340	11.34

Worked Example 1 — Calculating Density

```
Question: A solid metal block has a mass of 2 kg and a volume of 0.005 m³.
          Calculate its density.
```

```
Given:   m = 2 kg,   V = 0.005 m³
Formula: ρ = m / V
Working: ρ = 2 / 0.005
         = 400 kg/m³

Answer:  The density is 400 kg/m³ (or 0.4 g/cm³)

Interpretation: This material is less dense than water (1000 kg/m³),
                so a block of it would float in water.
```

Exam Tip: When calculating density, always check your units. If mass is in grams and volume in cm^3, you get g/cm^3. If mass is in kg and volume in m^3, you get kg/m^3. Examiners often ask you to convert between these units.

Required Practical 5: Measuring Density of Regular and Irregular Solids, and Liquids

Regular solids (e.g., rectangular block):

- Measure mass using a balance (in grams or kg)
- Measure dimensions (length, width, height) with a ruler or callipers
- Calculate volume: V = length × width × height
- Calculate density: ρ = m / V

Irregular solids (e.g., stone, key):

- Measure mass using a balance
- Use water displacement to find volume: place object in a measuring cylinder of water, measure the rise in water level
- Volume of object = final water level − initial water level
- Calculate density: ρ = m / V

Liquids:

- Measure mass of a known volume using a measuring cylinder and balance
- Calculate density: ρ = m / V

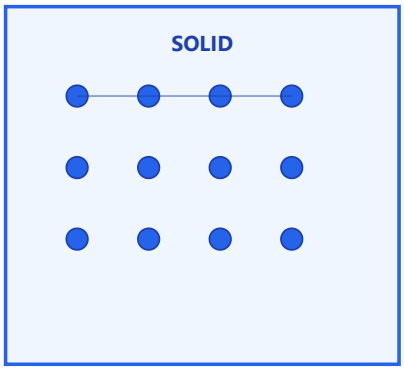
- Alternatively, use a hydrometer (a float that sinks to different depths depending on liquid density)

Exam Tip: In water displacement experiments, make sure the object is fully submerged. Also, wait for air bubbles to rise before reading the final water level—they would artificially increase the measured volume.

4.3.2 Changes of State and Internal Energy

Matter exists in three **states of matter**: solid, liquid, and gas. These states differ in the arrangement and motion of their particles.

Particle arrangement in the three states of matter

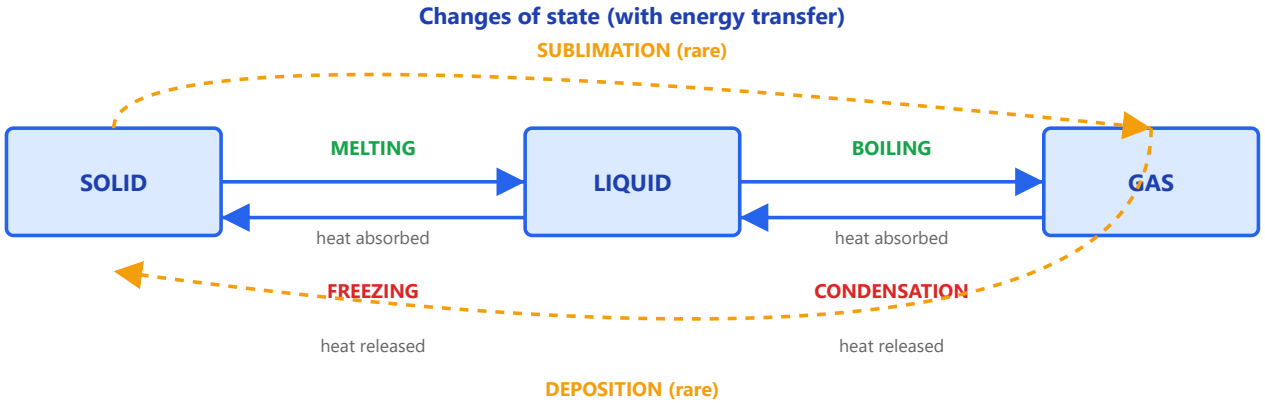


Particles: vibrate in fixed positions **Forces:** very strong **Spacing:** close, regular lattice LIQUID
Particles: slide past each other **Forces:** weaker than solids **Spacing:** close, but random GAS
Particles: fly apart rapidly **Forces:** almost none **Spacing:** far apart, random

Changes of State

A **change of state** occurs when matter converts between solid, liquid, and gas. The six main

changes are:



Melting: solid → liquid. Heat is absorbed; particles gain kinetic energy and break free from fixed positions. Temperature rises until melting point, then stays constant during melting.

Freezing: liquid → solid. Heat is released; particles slow down and lock into a regular lattice. Temperature falls until freezing point, then stays constant during freezing.

Boiling/Evaporation: liquid → gas. Heat is absorbed; particles escape into gas phase. Temperature stays constant during boiling until all liquid has turned to gas.

Condensation: gas → liquid. Heat is released; gas particles slow down and cluster together. Temperature stays constant during condensation.

Sublimation: solid → gas (without melting first). Occurs in a few substances, e.g., dry ice (solid CO_2) sublimates at −78 °C at normal atmospheric pressure.

Deposition: gas → solid. Occurs when a gas is cooled directly to a solid, e.g., frost forming from water vapour in air.

Key Facts About State Changes

Mass is conserved during a change of state. If 100 g of ice melts to form water, the water still has a mass of 100 g. No mass is gained or lost during melting, freezing, boiling, or condensation.

A change of state is a physical change, not a chemical change. The substance's identity does not change. Water (H_2O) remains water whether it is ice, liquid, or steam. No new substances are formed. The change is **reversible**—you can melt ice back to liquid water, then freeze it again to ice.

Exam Tip: Do not confuse changes of state with chemical reactions. Melting ice is physical; burning wood is chemical. In exams, always be clear: "This is a physical change because the substance's identity (e.g., H_2O) has not changed."

Internal Energy

Internal energy is the total energy stored in a substance due to the motion and arrangement of its particles. It is the sum of:

- **Kinetic energy:** energy due to the motion (vibration, sliding, flying) of particles
- **Potential energy:** energy stored in the bonds between particles (due to their separation and the forces between them)

How internal energy changes with temperature: Heating a substance increases the average kinetic energy of its particles, so they move faster (or vibrate more vigorously). This raises the internal energy and the temperature. Cooling decreases it.

How internal energy changes during a state change: During melting, boiling, or sublimation, the temperature stays constant, so the average kinetic energy does not change. However, the absorbed heat energy goes into breaking bonds between particles (increasing potential energy). After the state change, further heating again raises kinetic energy and temperature.

Exam Tip: A key exam concept: energy input during a state change does NOT raise temperature; it breaks bonds. Think of it as "energy being stored in the new arrangement of particles" rather than "making them move faster."

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4.3.3 Temperature Changes and Specific Heat Capacity

Specific heat capacity is the amount of energy required to raise the temperature of 1 kg of a substance by 1 °C (or equivalently, 1 K).

Symbol: `c`, measured in **J/(kg°C)** or **J/(kg·K)**

The relationship between energy, mass, temperature change, and specific heat capacity is:

$$\Delta E = m\,c\,\Delta\theta$$

where ΔE = energy transferred (J), m = mass (kg), c = specific heat capacity (J/(kg°C)), $\Delta\theta$ = temperature change (°C or K).

Rearranged forms:

- $m = \Delta E / (c\,\Delta\theta)$ — find mass if you know energy and temperature change
- $c = \Delta E / (m\,\Delta\theta)$ — find specific heat capacity
- $\Delta\theta = \Delta E / (m\,c)$ — find temperature change

Why Water Has a High Specific Heat Capacity

Water has a specific heat capacity of 4200 J/(kg°C), which is much higher than most other substances. This is because:

- Water molecules form strong hydrogen bonds with each other
- Breaking and reforming these bonds requires significant energy input when temperature changes
- A given amount of heat energy therefore produces a smaller temperature change in water than in other liquids

Practical consequences:

- Water is an excellent coolant (e.g., car radiators use water to absorb heat from the engine)

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- Water heats up and cools down slowly—coastal regions have milder climates because the ocean absorbs summer heat and releases it in winter, moderating temperature extremes
- A cup of water takes much longer to heat than, say, a cup of oil

Specific Heat Capacity Values (Common Substances)

Substance	Specific Heat Capacity (J/(kg°C))
Water	4200
Concrete	800
Aluminium	900
Iron	450
Copper	385

Worked Example 1 — Specific Heat Capacity Calculation

```
Question: Calculate the energy needed to raise the temperature of 0.5 kg
          of water from 20 °C to 100 °C.

Given:   m = 0.5 kg,   Δθ = 100 − 20 = 80 °C,   c = 4200 J/(kg°C)
Formula: ΔE = m c Δθ
Working: ΔE = 0.5 × 4200 × 80
         = 168 000 J
         = 168 kJ

Answer:  168 kJ (or 1.68 × 10⁵ J)

Interpretation: It takes 168 kilojoules of energy to heat half a litre
                of water from room temperature to boiling point.
```

Required Practical 1: Measuring Specific Heat Capacity of a Solid (or Liquid)

Typical procedure (using an electric heater):

1. Measure the mass of the solid (e.g., aluminium block) using a balance
2. Record the initial temperature of the solid with a thermometer
3. Pass a known current through a heater immersed in (or in contact with) the solid for a measured time
4. Record the temperature at regular intervals (e.g., every 30 seconds)
5. Repeat the heating and temperature measurement until a substantial temperature rise is achieved
6. Calculate the electrical energy supplied: $E = I^2 R t$ or $E = V I t$
7. Calculate the temperature change: $\Delta\theta$ = final temperature − initial temperature
8. Rearrange the specific heat equation: $c = \Delta E / (m \Delta\theta)$

Sources of error: Heat loss to the surroundings (the block is cooling while being heated), inaccurate thermometer readings, and uncertainty in measuring time or current.

> **Exam Tip:** In an exam, if asked about sources of error in a calorimetry experiment, always mention heat loss to the surroundings. This is the largest source of systematic error and would cause your calculated value of c to be too high (because some of the electrical energy went into heating the air, not the substance).

4.3.4 Changes of State and Specific Latent Heat

Specific latent heat is the energy required to change the state of 1 kg of a substance without a change in temperature.

Symbol: L, measured in **J/kg**

Formula: $E = m L$

where E = energy transferred (J), m = mass (kg), L = specific latent heat (J/kg).

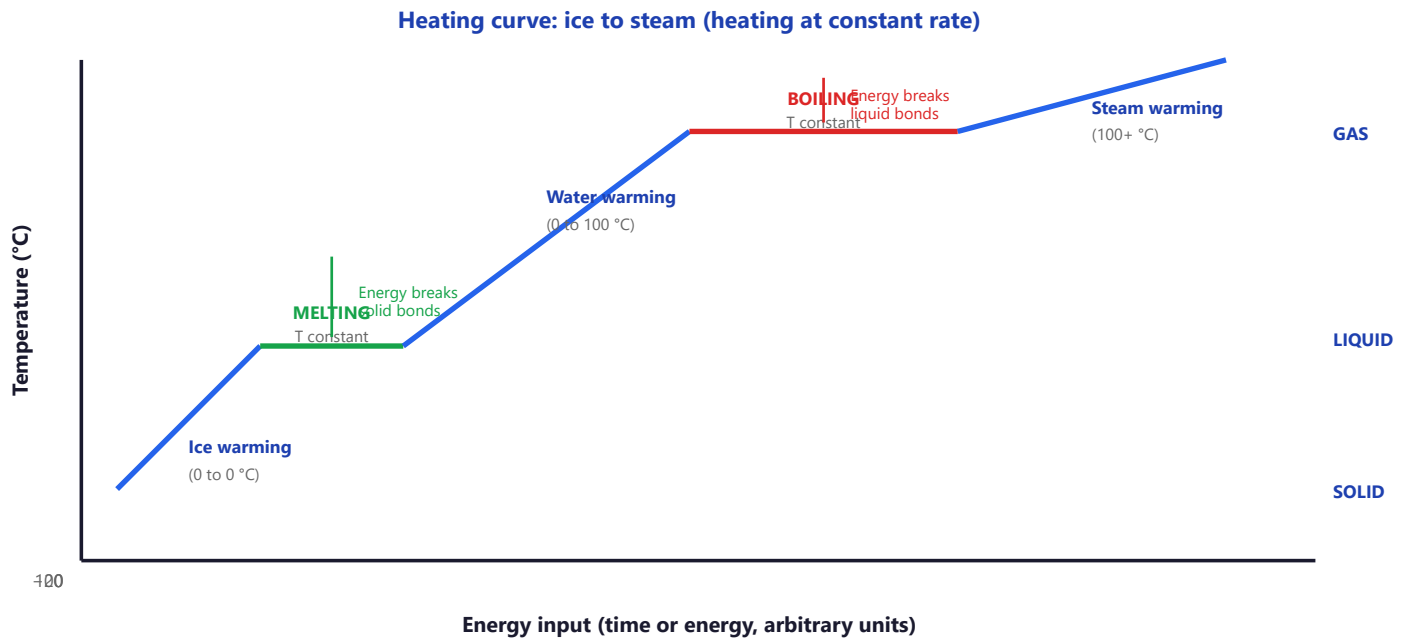
Two Types of Latent Heat

Specific latent heat of fusion (L_f): Energy needed to convert solid to liquid (or vice versa) at constant temperature. For water at 0 °C: L_f = 334 000 J/kg.

Specific latent heat of vaporisation (L_v): Energy needed to convert liquid to gas (or vice versa) at constant temperature. For water at 100 °C: L_v = 2 260 000 J/kg.

Why is $L_v > L_f$? During melting, particles need enough energy to escape their rigid lattice and slide past each other—bonds are partially broken. During boiling, particles must escape completely from the liquid and become widely spaced in the gas phase—bonds are almost entirely broken. Breaking more bonds requires more energy, so L_v is typically 5–10 times larger than L_f.

The Heating Curve: Temperature vs Energy Input



Key observations from the heating curve:

- **Sloped sections (ice warming, water warming, steam warming):** Temperature rises as energy is added. Slope shows the relationship: steep slope = low specific heat capacity (temperature rises quickly); shallow slope = high specific heat capacity (temperature rises slowly). Water's section has the steepest slope because water has the highest specific heat capacity.

- **Flat sections (melting plateau, boiling plateau):** Temperature stays constant even though energy is being supplied. All the energy goes into breaking bonds and changing the state, not into raising kinetic energy.
- **Length of plateaus:** The boiling plateau is much longer than the melting plateau because $L_v >> L_f$—it takes far more energy to boil all the water than to melt all the ice.

Worked Example 1 — Melting Ice

```
Question: How much energy is required to melt 0.2 kg of ice at 0 °C?

Given:   m = 0.2 kg,   L_f = 334 000 J/kg
Formula: E = m L_f
Working: E = 0.2 × 334 000
         = 66 800 J
         = 66.8 kJ

Answer:  66.8 kJ (or approximately 67 kJ)

Note:    This energy converts ice to liquid water at 0 °C. The
         temperature does NOT rise during melting.
```

Worked Example 2 — Boiling Water

```
Question: How much energy is needed to boil away 0.5 kg of water
          at 100 °C?

Given:   m = 0.5 kg,   L_v = 2 260 000 J/kg
Formula: E = m L_v
Working: E = 0.5 × 2 260 000
         = 1 130 000 J
         = 1.13 MJ

Answer:  1.13 MJ (or 1 130 kJ, or 1.13 × 10⁶ J)

Note:    Compare to the melting example: boiling half a kilogram of
         water requires about 17 times more energy than melting 0.2 kg
         of ice. This is because L_v >> L_f.
```

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Exam Tip: In exam questions, watch out for the difference between heating a substance and changing its state. Heating raises temperature (use $\Delta E = m c \Delta\theta$); changing state keeps temperature constant (use $E = m L$). Read carefully to identify which process is happening.

4.3.5 Particle Motion in Gases

In a gas, particles are widely spaced and move rapidly in random directions. They collide with each other and with container walls. The temperature of a gas is directly related to the average kinetic energy of its particles.

Relationship between temperature and particle motion:

- **Higher temperature:** Particles move faster; average kinetic energy is higher
- **Lower temperature:** Particles move slower; average kinetic energy is lower
- **Absolute zero (−273 °C or 0 K):** (Theoretical) particles would stop moving entirely; in reality, absolute zero cannot be reached

Gas Pressure

Pressure in a gas is caused by particles colliding with the container walls and exerting a force on them. Higher frequency and harder collisions = higher pressure.

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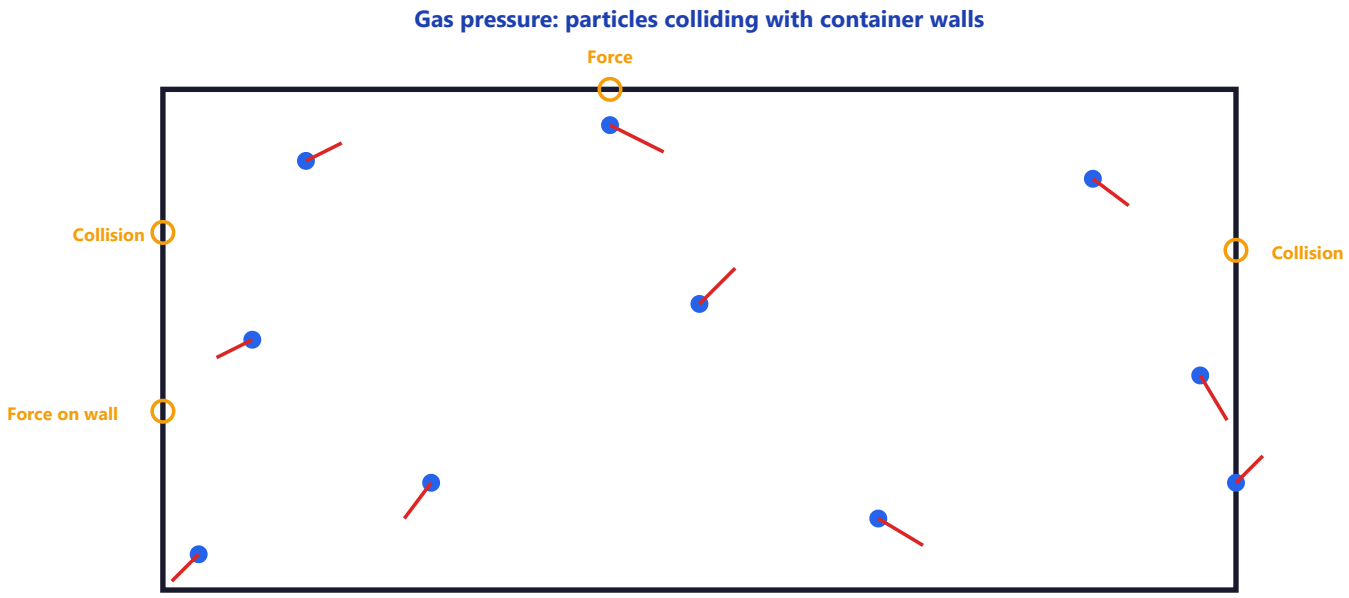


Pressure = total force / area = effect of particle collisions

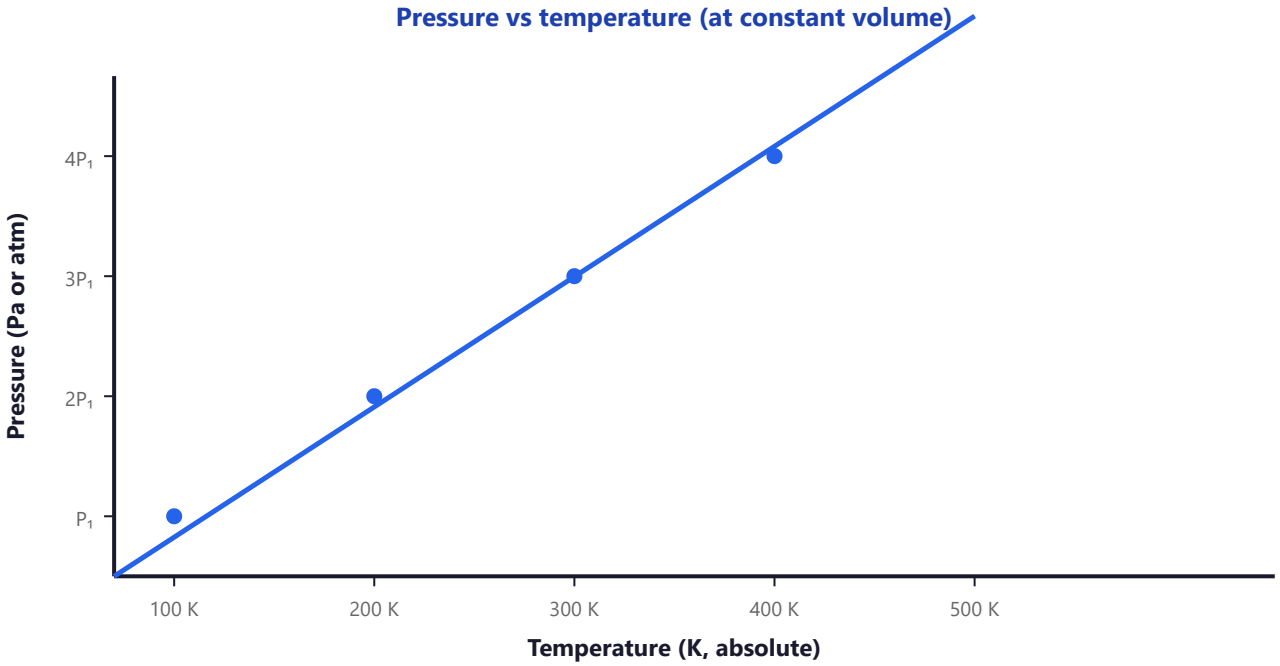
More collisions (higher T or lower V) → higher pressure

Effect of Temperature on Pressure (at Constant Volume)

If you heat a gas in a rigid container (constant volume):

- **Temperature increases:** Particles move faster
- **More frequent collisions:** Each particle hits the walls more often
- **Harder collisions:** Each impact transfers more momentum to the walls
- **Result:** Pressure increases

> At constant volume, **pressure is proportional to absolute temperature** (Gay-Lussac's Law in principle):
>
> `P ∝ T` (at constant V)
>
> or `P / T = constant`



Straight line through origin: P and T are directly proportional
If T doubles, P doubles (at constant volume)

Why does pressure increase with temperature? At a molecular level, higher temperature means faster particle motion. Faster particles hit the walls more frequently and with greater force, producing a higher pressure. This relationship only works if volume is held constant—if the container can expand, the volume change also affects pressure.

> **Exam Tip:** Use Kelvin (K), not Celsius, when working with gas laws. At 0 °C = 273 K. The proportionality $P \propto T$ only holds for absolute temperature. If you use Celsius, the relationship breaks down.

4.3.6 Increasing the Pressure of a Gas (Higher / Triple Science)

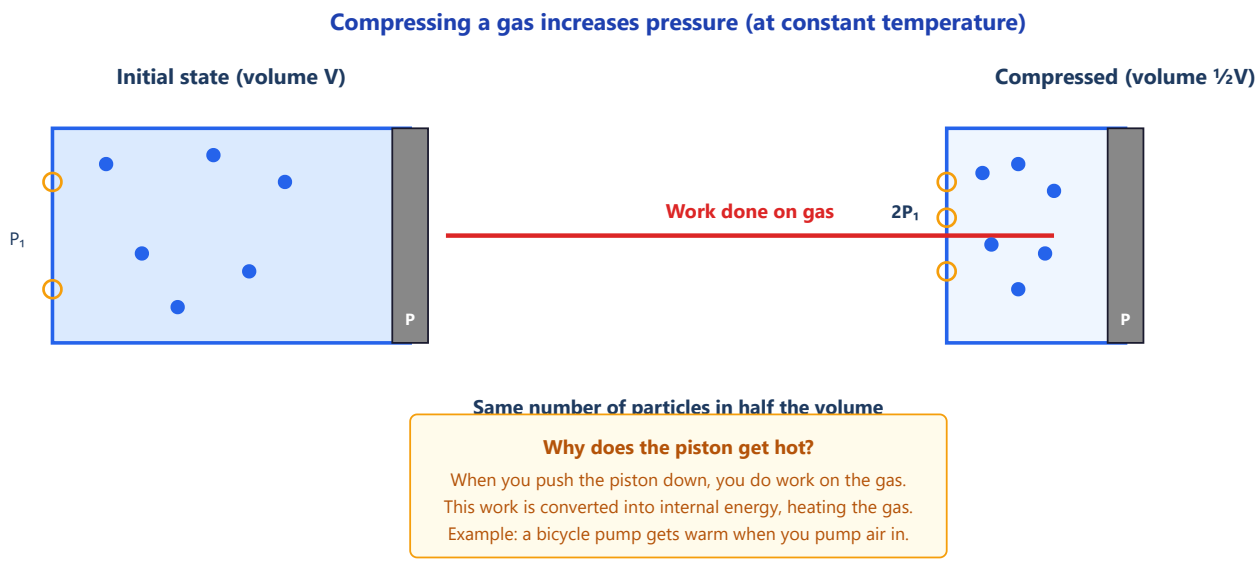
Compression at Constant Temperature

When you compress a gas by reducing its volume (e.g., pushing a piston into a syringe):

- **Volume decreases:** Particles are squeezed into a smaller space
- **Particle density increases:** Same number of particles in less volume
- **More frequent collisions:** Particles hit the walls more often (per unit area per unit time)

- **Result:** Pressure increases

> **Boyle's Law (stated in principle):** At constant temperature, pressure is inversely proportional to volume:
>
> `P ∝ 1/V` (at constant T)
>
> or `P V = constant`



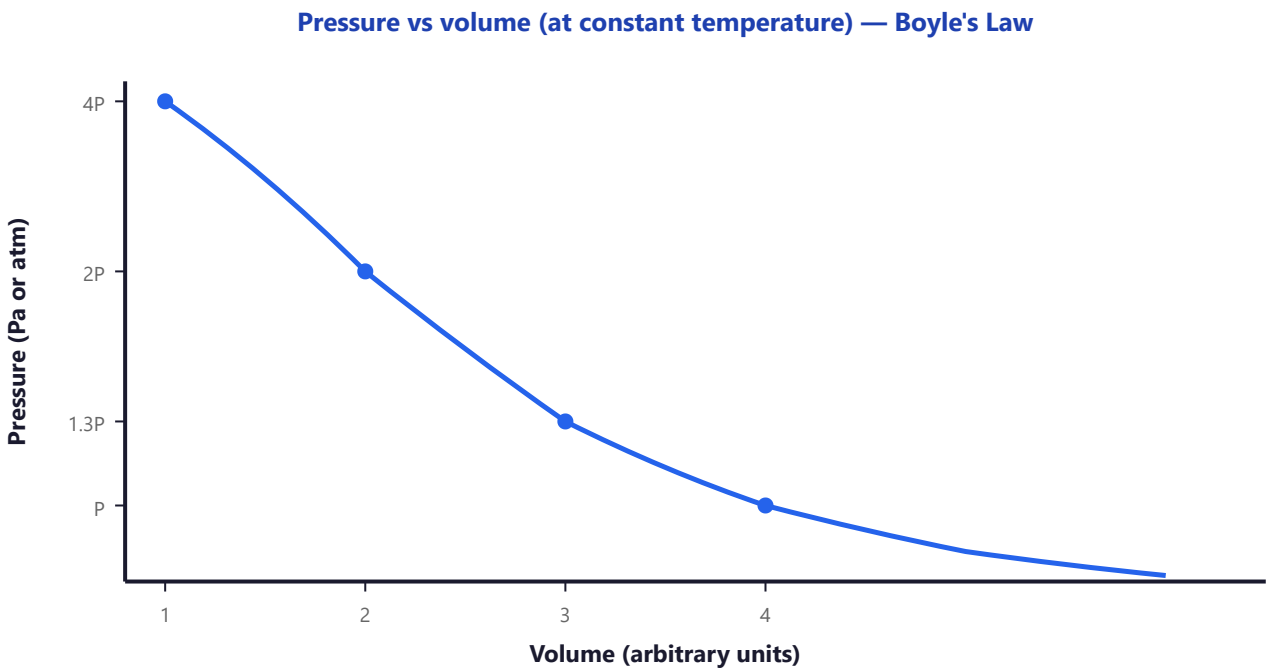
Practical Example: The Bicycle Pump

When you push down on a bicycle pump handle:

1. **Volume decreases:** The air chamber shrinks as the piston moves down
2. **Pressure increases:** The same number of air molecules are squeezed into a smaller space, creating more collisions and higher pressure
3. **Work is done on the gas:** Your muscles transfer energy as you push the piston
4. **Internal energy increases:** This energy goes into increasing the kinetic energy of the gas molecules, raising its temperature
5. **The pump gets warm:** You can feel this heat—the barrel of the pump becomes noticeably warmer after several strokes

Work done on a gas during compression: When you compress a gas, you transfer energy to it by doing work. This energy becomes internal energy (kinetic and potential energy of particles). If no heat is lost to the surroundings, the temperature of the gas rises.

Exam Tip: Examiners love the bicycle pump example because it links three concepts: (1) volume decrease → pressure increase, (2) work is transferred as energy, (3) internal energy increases → temperature increases. If asked "why does a bicycle pump get warm?" you should mention all three.



Hyperbolic curve: P and V are inversely proportional
If V is halved, P doubles (at constant temperature)

Reading the graph: The curve is hyperbolic (not straight). When volume is halved, pressure doubles. When volume is tripled, pressure becomes one-third. This inverse relationship—PV = constant—is Boyle's Law.

Exam Tip: Boyle's Law graph is curved, not straight. The curve shows that as volume increases, pressure decreases, but the relationship is not linear. At very large volumes, pressure changes are smaller. This is an important distinction from the P-T graph (which is linear).

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